



A RegTech Framework for Privacy, Resilience, and Regulatory Readiness





Executive Summary

This whitepaper by the Stablecoin Standard (SCS) explores the evolving RegTech landscape supporting financial crime prevention in blockchain-based financial systems. It highlights how emerging technologies—AI, APIs, digital identity tools, and blockchain analytics—are enabling real-time, privacy-aware compliance.

Traditional compliance systems are arguably misaligned with high-velocity digital finance. Fragmented tools and overreliance on personal data impede both effectiveness and trust. In response, many RegTech vendors are building modular, API-native and blockchain architectures that support dynamic oversight while limiting data exposure.

SCS introduces a functional taxonomy—Prevent, Detect, Trace, Protect—to categorize RegTech solutions across the financial crime lifecycle. The research shows growing momentum, but limited convergence: no vendor yet offers a full-stack compliance solution, and AML tools are often conflated with cybersecurity infrastructure, despite serving distinct functions.

The paper advocates for a shift to embedded compliance architectures that are modular, interoperable, and programmable. It also advocates that regulations should evolve to recognize blockchain-native tools that support fraud prevention without centralizing personal data—enhancing financial integrity, privacy, and regulatory efficiency.



Key Takeaways:

- RegTech is enabling continuous, risk-based compliance aligned with blockchain transparency.
- AML/KYC and cybersecurity must be treated as complementary but distinct controls.
- Compliance architecture should prioritize interoperability, transparency, and data minimization.
- Public-private coordination is essential to modernize policy frameworks and ensure regulatory readiness.

This paper is not a vendor ranking or comprehensive market scan. It is a strategic blueprint to inform regulators, technologists, and institutions shaping the next generation of digital compliance systems.

Introduction

The Stablecoin Standard (SCS) conducted targeted research into regulatory technology (RegTech) tools relevant to financial crime prevention in blockchain-based systems. This work focuses on technologies that align with evolving priorities around transparency, traceability, and real-time supervision—including tools for transaction monitoring, digital identity verification, blockchain analytics, and cybersecurity.

Traditional compliance architectures—designed for closed, intermediary-based systems—struggle to adapt to decentralized networks and permissionless infrastructure. In response, RegTech vendors are developing modular, API-native tools that offer more adaptive, scalable, and automation-friendly compliance capabilities for the digital asset sector.



The research categorizes tools by functional outcomes—Prevent, Detect, Trace, Protect—which mirror the financial crime lifecycle. However, an important distinction is often overlooked: AML/KYC and blockchain analytics are not substitutes for cybersecurity. The former identify and monitor users and transactions; the latter protect the infrastructure itself against fraud, theft, and technical failure. Both are essential to system integrity, but they serve fundamentally different functions and should not be conflated.

While some vendors span multiple categories, most deliver narrowly scoped solutions. The result is momentum, but not yet convergence: integration across these functional layers remains limited, and no single vendor currently provides an end-to-end compliance stack.

This research also highlights an emerging policy tension: as institutions seek greater visibility, some vendors continue to rely on expansive personal data collection. SCS believes that modern compliance infrastructure must evolve to minimize unnecessary data exposure—leveraging blockchain transparency and privacy-preserving technologies to enable risk-based oversight without compromising individual privacy.

This whitepaper is not a comprehensive market survey. It is an early step in documenting how RegTech innovation can support more effective, privacy-aware compliance in digital finance. SCS welcomes collaboration with regulators, policymakers, and vendors to refine these insights and help shape a more resilient and accountable financial system.

Methodology: Research Approach and Framework

SCS conducted a review of select RegTech vendors operating across the financial-crime lifecycle. The research focused on firms whose technologies align with priorities for transparency, data security, and illicit-finance prevention in digital and traditional financial systems. Sources included vendor documentation, technical whitepapers, regulatory filings, and interviews with subject-matter experts. Each solution was evaluated for technological maturity, interoperability with public blockchains and digital-asset infrastructure, and relevance to emerging policy frameworks.

The solutions have been classified by the four primary outcomes they deliver in the financial-crime lifecycle: Prevent, Detect, Trace, Protect. This taxonomy mirrors the functional stages of the financial-crime lifecycle—from identity verification and onboarding to investigation and infrastructure defense.

- Solutions that help prevent financial crime focus on keeping bad actors out of the system altogether - verifying identities, credentials, and eligibility before onboarding to ensure only legitimate users gain access.
- In contrast, detect solutions monitor activity in real time to identify suspicious or anomalous behavior across payments, accounts, and transactions, allowing institutions to respond before harm occurs.
- Trace solutions come into play after a signal or incident, mapping funds, entities, and relationships across blockchain and traditional networks to uncover illicit flows and support investigations.
- Finally, protect solutions secure the underlying infrastructure itself - safeguarding wallets, custody systems, and transaction layers to prevent theft, exploits, and misuse before they can take place.

Category	Purpose	Primary Capabilities	Sample Vendors
Prevent 	Keep bad actors out of the system before they transact.	Identity verification, KYC/KYB automation, credential attestation, and fraud prevention.	Persona, Humanity Protocol, Bluprynt, Sphinx, Blockaid, Predicate
Detect 	Monitor activity in real time to identify suspicious or anomalous behavior.	Transaction monitoring, behavioral analytics, sanctions and PEP screening.	Flagright, Unit21, Castellum.AI, ThetaRay, SafetyKit
Trace 	Map illicit flows after a signal or incident to support investigations.	Blockchain forensics, entity clustering, fund-flow visualization.	TRM Labs, Inca Digital
Protect 	Secure the financial infrastructure and prevent exploitation or theft.	Wallet security, custody monitoring, fraud prevention for payment and DeFi systems.	Fireblocks, Hypernative

Table 1: See the Appendix for more information

Context: A New Architecture for Financial Crime Prevention



Legacy compliance frameworks were built for closed-loop systems with intermediaries, static rules, and periodic reviews. These models struggle to keep pace with the speed, complexity, and fragmentation of financial activity in digital markets. As transaction volumes rise and span multiple jurisdictions, institutions increasingly rely on broad personal data collection to offset weak signals from outdated, siloed systems—raising both operational costs and privacy concerns without meaningfully improving risk detection.

Modern RegTech innovation presents a clear opportunity to shift from reactive, manual processes to proactive, data-driven oversight. Emerging tools—including AI-enabled monitoring, blockchain analytics, verifiable digital identity, and API-native architectures—enable continuous compliance aligned with the principles of transparency, auditability, and risk-based supervision. These technologies offer dynamic capabilities that can scale across the unique characteristics of digital assets and public blockchains.

Public blockchain data, in particular, introduces a foundational shift in what's possible: it supports traceability without requiring institutions to centralize sensitive user data. Instead of relying solely on static KYC files, institutions can leverage real-time on-chain intelligence and privacy-preserving verification to meet compliance objectives. This evolution supports a more efficient compliance posture—one that protects both financial integrity and user privacy.

Still, confusion remains between key functional areas. AML/KYC solutions and blockchain analytics—focused on identifying users and tracing transactions—are often incorrectly treated as substitutes for cybersecurity. In reality, cybersecurity protects the infrastructure itself from theft, fraud, and exploitation. Both domains are essential but distinct. Institutions must recognize and invest in these layers separately.



Current RegTech offerings remain fragmented. Few vendors span the full financial crime lifecycle, and most specialize in discrete stages: prevention, detection, tracing, or protection. This creates integration challenges, but also opportunities. A cohesive compliance architecture—built on modular tools and privacy-aware standards—can deliver stronger results than siloed controls.

The next generation of compliance systems will depend not just on new technologies, but on thoughtful design: architectures that support interoperability, minimize data exposure, and reflect the programmability of digital finance. This vision aligns with emerging frameworks such as [*"Trust Without Intermediaries": A Programmable Risk Management Framework for the Future*](#) which advocate for programmable risk architectures that embed trust at the protocol level rather than relying on traditional intermediaries. As programmable finance matures, the ability to engineer risk controls directly into infrastructure—while preserving privacy and resilience—will become foundational to modern compliance design.

Core Technologies Driving RegTech Innovation

A. Artificial intelligence and machine learning

Artificial intelligence (AI) and machine learning (ML) have become central to modern financial crime prevention. Across the vendor landscape, AI is being used for anomaly detection, behavioral analytics, and dynamic risk scoring. These systems process vast volumes of structured and unstructured data to identify subtle deviations from expected activity—detecting emerging threats that traditional rule-based models often miss.



Research indicates that vendors are designing and deploying adaptive AI models that continuously learn from transaction data and feedback loops. This allows for the potential of real-time risk scoring and model recalibration as new typologies of fraud or money laundering appear. AI-driven detection substantially reduces false positives by distinguishing normal transaction behavior from genuinely suspicious activity. Applications range from transaction monitoring and sanctions screening to predictive analytics for fraud prevention and customer-risk management.

B. API-first compliance infrastructure

Application Programming Interfaces (APIs) have become the backbone of modern compliance frameworks. Rather than relying on closed or siloed systems, most RegTech vendors now design API-native solutions that prioritize interoperability and data portability. This architecture enables seamless integration with banking systems, fintech platforms, and digital-asset networks, creating a unified compliance layer that can communicate securely across multiple environments.

These API-based designs provide functional advantages. They enable real-time reporting, dynamic risk assessment, and modular compliance architectures that can be configured or expanded as regulatory requirements evolve. By embedding compliance directly into transaction and onboarding workflows, institutions can automate monitoring and adapt rapidly to new jurisdictions or asset classes. The result is a more flexible, cost-efficient framework for regulatory compliance - one that reduces fragmentation while improving both oversight and operational speed. However, API integrations can also facilitate the over-collection or oversharing of personal data when institutions receive broad data access simply because it is technically easy to do so. A clear regulatory framework that enforces purpose-limitation and data-minimization can preserve API efficiency while ensuring privacy-by-design.



C. Blockchain analytics and on-chain intelligence

Blockchain analytics has emerged as a critical component of financial-crime prevention in the digital-asset ecosystem. New analytical frameworks can monitor public ledgers, identify relationships between wallets, and assign risk scores based on behavioral and transactional patterns. These tools help bridge the gap between on-chain transparency and regulatory accountability.

RegTech vendors are now able to cluster wallet addresses, attribute them to entities, and link on-chain activity to off-chain identifiers. By combining blockchain intelligence with traditional KYC and AML data, institutions can trace illicit flows, enforce sanctions, and detect ransomware or terrorist-financing activity with unprecedented precision. Notably, while AML/KYC and blockchain analytics identify and trace suspicious actors, they do not prevent unauthorized access or theft. Cybersecurity tools such as wallet protection, custody monitoring, and network security are necessary to actively safeguard the infrastructure that supports these transactions.

This convergence of blockchain analytics and traditional compliance infrastructure enhances the ability of regulators and institutions to track funds across both fiat and digital networks while still preserving privacy.

D. Digital identity and verification technologies

Digital identity innovation is reshaping how institutions verify and onboard users. The shift toward verifiable, user-centric identity systems—often incorporating biometric verification, decentralized identifiers (DIDs), and proof-of-humanity



mechanisms—enhances both accuracy and privacy. These technologies allow legitimate users to prove who they are without exposing unnecessary personal information, reducing the risk of identity theft and data breaches.

The select review of RegTech vendors highlights the adoption of biometric and cryptographic verification methods that validate credentials across jurisdictions. Such approaches support financial inclusion by enabling secure participation for users without traditional identity documents, while strengthening compliance and enhancing consumer privacy through minimized data collection and user-controlled verification. The next stage of development points toward interoperable digital-identity standards that can operate seamlessly across borders—establishing a foundation for secure, privacy-preserving financial access in the digital economy.

Redefining Compliance Architecture for Digital Finance

The potential convergence of AI, APIs, blockchain analytics, and digital identity unlocks a critical opportunity: reimagining compliance as a continuous, embedded architecture rather than a fragmented set of tools.

Each technology has the potential to strengthen a different layer of the compliance stack—intelligence, traceability, identity assurance, and interoperability. When combined, they enable onboarding, monitoring, investigations, and enforcement to operate in sync, reducing fragmentation and manual overhead while enhancing privacy and systemic trust.

A modern compliance architecture should be modular, composable, and programmable—designed to embed enforcement logic directly into digital workflows.



By prioritizing interoperability and data minimization from the outset, such systems can scale across jurisdictions while aligning with privacy-preserving design principles.

This approach aligns with models like *Trust Without Intermediaries: A Programmable Risk Management Framework for the Future*, which advocate for embedding risk controls at the infrastructure level. These systems do not eliminate oversight; they make it more effective by ensuring accountability is built into the protocol itself.

To realize this shift, implementing regulations must evolve to support compliance models that leverage public blockchain transparency while minimizing reliance on expansive personal data collection. This modern architecture enhances fraud prevention, reinforces systemic trust, and supports supervisory models that are dynamic, privacy-aware, and digitally native.

Conclusion: RegTech Innovation Is Shaping the Future of Financial Compliance

Regulatory technology is reshaping how institutions detect, investigate, and prevent financial crime. The tools reviewed in this research—spanning AML/KYC, transaction monitoring, blockchain analytics, and cybersecurity—demonstrate how modular, interoperable infrastructure can advance both compliance efficiency and system resilience.



Three principles should guide the next generation of compliance architecture:

- Interoperability – to ensure that risk-relevant information can move securely and effectively across systems and jurisdictions.
- Transparency – to enable explainable, auditable compliance processes aligned with regulatory expectations.
- Data minimization – to reduce reliance on personal data while maintaining high-integrity monitoring and enforcement.

These priorities are especially critical as compliance shifts from static rule-based checks to real-time, risk-based oversight. However, while RegTech innovation is accelerating, the market remains fragmented. No single vendor spans all functions—AML/KYC tools are not cybersecurity solutions, and infrastructure protection must be prioritized alongside user surveillance.

Policy frameworks must evolve accordingly. Rather than retrofitting analog compliance rules to digital systems, regulators have an opportunity to promote standards that reflect the unique transparency, automation, and composability of public blockchains. Modern RegTech offers a path toward supervisory models that are continuous, embedded, and privacy-preserving.

To fully realize this potential, sustained public-private collaboration will be essential. SCS invites further engagement with technologists, policymakers, and regulators to advance shared goals around financial integrity, user protection, and responsible innovation.

Appendix - Select RegTech Vendors



Prevent

Vendor	 persona
Description	Helps businesses detect and prevent fraud and money laundering through multi-layered identity verification, watchlist screening, and link analysis. By authenticating users and automating KYC/KYB checks, it ensures only legitimate entities can access financial systems
Tech used	- AI to detect spoofs/deepfakes and anomalous behavior. - APIs / OIDC to embed verification and risk screening into onboarding and access flows - Digital identity verification across the lifecycle; aligned with NIST IAL - Privacy-enhancing approach with passive signals, minimal data exposure.
Resources	Persona homepage Variety of ebooks published by Persona

Vendor	 humanity protocol
Description	Prevents illicit finance by verifying real humans using palm-based biometrics that prove personhood without exposing personal data. This “proof of human” identity layer strengthens transparency and accountability across digital and financial ecosystems, reducing fraud and identity-based money-laundering risks.
Tech used	- Digital Identity Verification Biometric verification to block synthetic and/or duplicate accounts, zero-knowledge proofs enable verification without collecting personal data, Blockchain provides auditable record of verified users. - Privacy-Enhancing Tools Implements a zero-knowledge proof (ZKP)-based identity layer where users verify humanity without exposing personal data. Biometric palm scans are processed locally into non-reversible templates, and only cryptographic attestations are stored or shared.
Resources	Proof of Humanity Humanity Protocol: The Identity Solution Going after Meta - Delphi Digital

Vendor	 BLUPRYNT
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Description	Verifies token issuers through its KYI protocol, which links wallets and token contracts to verified legal entities, preventing fraudulent or anonymous token issuance. By combining SmartDocs, PolicyEngine, and on-chain attestations, the platform embeds compliance logic directly into digital assets, enabling real-time verification, transfer restrictions, and auditability. These mechanisms enhance transparency and trust across digital asset markets.
Tech used	- AI Employs AI-driven automation through the Bluprynt PolicyEngine (BPE) to interpret regulatory requirements and dynamically apply rule sets across token issuance and transfers. This system streamlines compliance and disclosure workflows, reducing manual intervention while ensuring real-time regulatory alignment. - API Provides on-and-off-chain APIs allowing exchanges and wallets to auto verify KYI status in real time. These APIs embed policy enforcement directly into token transfers, ensuring that only verified and compliant assets circulate. - Digital Identity Verification Uses blockchain attestations and SmartDocs to publicly verify token provenance, linking legal entities to on-chain activity through KYI credentials. This cryptographic identity framework authenticates issuers, prevents counterfeit or unauthorized token creation, and reinforces transparency across digital-asset ecosystems. - Blockchain Transaction Monitoring Applies SmartDocs and PolicyEngine enforcement to embed immutable compliance rules and audit trails directly on-chain, enabling continuous monitoring, traceability, and proactive

	detection of illicit token flows.
Resources	Bluprynt Partners with Chainlink to Deliver Automated Web3 Compliance for Digital Assets

Vendor	Sphinx*
Description	Identifies and reduces risks by detecting suspicious accounts and hidden connections. Supports early detection, transparency, and compliance while lowering false positives.
Tech used	AI agents automate KYC/AML onboarding, graph-based mapping uncovers hidden entity relationships, and automated screening/reporting tools flag risks from sanctions, PEP's and adverse media in real-time.
Resources	Sphinx home page Sphinx whitepapers

Vendor	Blockaid
Description	Provides an end-to-end solution that can help any company building in the space keep their users safe - including tools like smart contract monitoring, transaction simulation, dAPP

	scanning, token security tools and on-chain and off-chain threat hunting capabilities. It is able to provide strong on-chain security while avoiding unnecessary personal-data collection by relying on privacy-preserving techniques such as transaction simulation, behavioral analysis, and contract-level threat detection rather than gathering user-level identifiers.
Tech used	-AI Uses machine learning models, heuristic engines, clustering algorithms, and predictive risk scoring to identify malicious transactions, contracts, websites and wallet behavior in real time. These systems analyze both on-chain and off-chain data to detect scams, drainer scripts, phishing attempts and abnormal smart-contract behavior. -API Transaction simulation APIs, threat-detection APIs and webhook alert APIs that allow wallets, exchanges and dApps to embed its security engine directly into users flows. Blockaid simulates transactions before execution, flags malicious interactions, and surfaces real-time alerts to developers and users. -Digital Identity Verification Verifies address reputation, contract identity, dApp authenticity, and on-chain behavioral identity to legitimize/de-legitimize wallets, contracts and interfaces. Contrary to traditional KYC, it evaluates identities based on activity, behavioral patterns and signature identity. -Privacy Enhancing Tools Uses privacy-preserving transaction simulation, minimal-data security models and off-chain phishing detection. It determines whether a transaction is safe without collecting personal information or requiring access to private keys.

Resources	Blog
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Vendor	 PREDICATE
Description	Brings institutional-grade compliance, regulatory readiness, and business-policy enforcement to blockchain apps, enabling regulated finance, real-world asset tokenization, or institutionally compliant DeFi with minimal friction
Tech used	-API REST API that evaluates transactions against predefined compliance policies before they execute on-chain. Users send transaction data to the API, which then performs checks such as AML/sanctions screening, geolocation restrictions, collateral thresholds, and rate limits. Then, the API returns a cryptographic attestation confirming whether a transaction is compliant. -Digital Identity Verification Predicate Identity - Users complete identity verification through a link or QR code, and the system issues a verified-identity verification that can be used for all future on-chain interactions. This allows dApps to enforce identity-based rules. -Privacy Enhancing Tools Supports zero-knowledge identity verifications through integrations with privacy-focused providers such as human.tech. These tools allow users to prove attributes such as identity, uniqueness or eligibility without providing personal information to the smart contract.

Resources	Whitepaper
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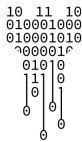
Detect

Vendor	 Flagrigh
Description	Detects and prevents illicit flows in real time through an integrated compliance platform. Consolidates alerts into holistic investigation cases with AI-generated insights and contextual summaries. Continuously assesses customer and merchant risk, adjusting monitoring intensity to behavioral changes. Screens against global sanctions, PEP, and adverse-media data to reduce exposure and false positives. Enhances transparency, traceability, and response speed, enabling proactive disruption of money laundering and fraud.
Tech used	- Generative AI Generative AI and AI-native AML investigations combine natural language queries, automated insight summaries, and real-time analytics to detect suspicious activity across vast datasets. They reduce false positives by up to 93%, continuously learn from new typologies of money laundering and fraud, and adapt in real time to emerging risks. - API API-driven real-time and batch transaction monitoring provides millisecond oversight of transactions, instantly flagging

	suspicious activity. Seamless integration across systems ensures full traceability and rapid response to illicit flows - Digital identity Verification AI-powered KYC and KYB solutions dynamically assess individual and merchant risk during onboarding and throughout the customer lifecycle. By combining behavioral analytics and risk scoring, they identify high-risk entities early, strengthen customer due diligence, and enable risk-based monitoring that prevents the misuse of accounts for illicit finance. - Blockchain Transaction Monitoring Generative AI-enhanced rule simulation and risk-based monitoring trace funds across fiat and blockchain networks, detecting anomalies and layering patterns in cross-border and decentralized transactions. - Privacy-Enhancing Tools Comprehensive data sources ensure accuracy and transparency in sanctions and PEPs screening, reducing redundant personal data proceeding. Flexible configuration minimizes false positives through customizable screening options, allowing institutions to focus only on relevant alerts while maintaining compliance with data protection standards.
Resources	Whitepaper: Leadership in Finance: Integrating Compliance and Technological Innovation White paper Real-Time Risk Scoring in AML Compliance: Flagright's Modern Approach

Vendor	
Description	Provides an AI-driven, no-code compliance platform that detects and prevents fraud, money laundering and other financial crimes. Its integrated and configurable system streamlines investigations, centralizes data for transparent reporting and reduces false positives.
Tech used	- AI Uses machine learning and behavior-based analytics to detect suspicious transaction patterns, identify evolving fraud typologies, and block high-risk payments in real time. By learning from feedback loops within rule testing, the system continuously improves accuracy, reducing false positives and enabling proactive disruption of money mule activity. - No-code API Provides a no-code, API-first infrastructure that connects transaction monitoring, fraud detection , and casemanagement in real time. Supports configurable rules and workflows, allowing neobanks to tailor compliance processes, automate SAR filings, and streamline investigations with minimal engineering effort. - Digital Identity Verification Automates KYC/KYB during onboarding through integrated data sources and dynamic risk scoring. It prevents account-opening fraud. - Privacy-Enhancing Tools Implement granular access controls, automated workflows, and audit-ready reporting to ensure sensitive customer data is used securely and transparently. Reduces manual handling of

	personal data, lowering exposure risk.
Resources	Unit 21 whitepaper

Vendor	 Castellum.AI
Description	Automates financial crime detection with real-time global risk data and explainable AI, improving screening accuracy, investigative speed, and compliance efficiency. Its adaptive platform integrates entity intelligence, sanctions data, and behavioral analytics across jurisdictions to uncover hidden risk relationships and prevent misuse of financial systems. This unified approach strengthens regulatory compliance and enables institutions to detect and disrupt money-laundering activity on scale.
Tech used	- AI Employs agentic and explainable AI that can plan, act, and adapt autonomously in AML workflows. It analyzes ambiguous data, resolves fuzzy matches, and pivots during investigations while maintaining traceable decision logic. This adaptive intelligence identifies complex laundering networks missed by static rule-based systems, reducing alert fatigue and improving response speed. - API Built on a modular, API-native architecture that enables seamless interoperability between AI agents, data streams, and compliance workflows in real time. - Digital Identity Verification

	Integrates entity- and ownership-level intelligence by using autonomous agents to interpret documents, sanctions, adverse-media and beneficial-owner data. Through cross-jurisdictional linkage of identity and entity attributes, the platform prevents account-opening fraud, blocks money-mule access, and strengthens defenses against laundering schemes that exploit layered ownership. - Privacy Enhancing Tools Implements explainable AI that ensures full auditability, reproducibility, and justification for every decision, eliminating “black-box” opacity in compliance workflows. Every model output is logged and versioned—from input data and triggered logic paths to scoring components and decision outcomes—allowing any action to be reviewed and validated retroactively. This transparency empowers compliance professionals, not data scientists, to understand and defend AI-driven results, aligning with global standards for model risk management and regulatory accountability.
Resources	<u>Agentic AI: How Financial Compliance Is Outgrowing Rules-Based Automation</u> <u>Explainable AI: Compliance Without the Black Box</u> <u>Modernizing AML Compliance in Credit Unions: Expert Panel — Castellum.AI</u>

Vendor	 THETARAY
Description	Uses AI-driven transaction monitoring to detect hidden financial crimes in cross-border payments, helping banks and fintechs block suspicious activity in real time. Analyzes vast global payment data to uncover anomalies and complex money-laundering patterns that traditional rules-based systems miss. Provides explainable, regulator-aligned insights that enable institutions to distinguish legitimate transactions from illicit flows and maintain compliance across jurisdictions
Tech used	- Cognitive AI Hybrid Explainable AI identifies complex fraud patterns, unsupervised anomaly detection flags unusual transactions, graph analytics uncovers hidden connections between entities in real-time. Unlike opaque black-box AI, explainable architecture provides feature-level transparency for every alert, reducing false positives to 1% and aligning with OCC, FinCEN, and FFIEC model-risk expectations. - APIs Delivers cloud-native SaaS APIs that integrate AI-driven screening, transaction monitoring, and sanctions filtering directly into existing compliance workflows without system redevelopment. - Digital Identity Verification Applies AI-enhanced screening across global KYC and sanctions data sets to verify customer and counterparty identities in multiple languages and scripts. - Privacy-Enhancing Tools

	Implements Explainable-AI governance aligned with NIST AI risk management and Model Risk Management frameworks, ensuring every decision is traceable, auditable, and compliant. This transparency removes black-box opacity while upholding data protection and operational efficiency.
Resources	ThetaRay homepage Building Trust Through Explainability in Screening: A Model Risk Management Perspective AI: The Future of Financial Crime Detection

Vendor	 SafetyKit
Description	Uses AI agents to monitor and review platform activity at scale. By automatically detecting fraud, transaction laundering, illicit-merchant behavior and content violations, the system prevents bad actors from exploiting payment and marketplace platforms. Ensures continuous risk enforcement and compliance before funds move through modular APIs that automate onboarding and transaction screening.
Tech used	- AI Employs AI agents to handle complex workflows across risk, compliance and safety. Uses multi-modal analysis of images, videos, and text to detect seller scams, fake listings, and off-platform payment redirects before transactions complete. Graph-based analytics map coordinated fraud networks across

	merchant clusters, identifying organized abuse missed by rule-based systems and reducing chargebacks and dispute risks. - API Designed a modular API to automate onboarding reviews, seller risk screening, product safety checks and transaction monitoring. These processes enable pre-transaction screening and risk enforcement within existing payment and marketplace workflows, preventing fraudulent activity before funds move.
Resources	Agentic Compliance Fraud Prevention with AI



Trace

Vendor	
Description	Detects and disrupts illicit financial activity by tracing and monitoring blockchain transactions. Maps wallet addresses, tracking flow across blockchains, and identifying links to scams, terrorist financing, and money laundering schemes. Combines blockchain intelligence with off-chain data and regulatory

	context to deliver actionable insights that help institutions freeze or recover assets, strengthen AML compliance, and safeguard financial integrity.
Tech used	- AI Employs AI-driven analytics to detect suspicious on-chain and fiat-linked patterns in real time. Machine-learning models identify anomalous wallet behavior, flag illicit activity, and reduce false positives by continuously learning from new transaction typologies and regulatory feedback. - API Provides secure API connectivity between banks, wallets, and blockchain nodes to enable automated screening and risk scoring before stablecoin transactions are processed. Integration with the core banking systems and issuer platforms allows real-time mint/redeem visibility and rapid response to suspicious flows. - Digital Identity Verification Applies comprehensive KYC and customer-due-diligence processes that verify user and wallet ownership at onboarding and throughout the lifecycle. AI-enhanced screening checks sanctions, politically exposed persons (PEPs), and high-risk jurisdictions to prevent misuse of accounts for illicit finance, while sanctions and OFAC checks ensure compliance, and entity profiling links on-chain behavior to real-world actors. - Blockchain Transaction Monitoring Detects illicit activity patterns, wallet screening identifies high-risk addresses. Provides real-time alerts on wallet movements, cross-chain and cross-exchange flows, and concentration risks to support proactive freezing, seizure, and recovery of illicit assets.

	- Privacy-Enhancing Tools Provides real-time alerts on wallet movements, cross-chain and cross-exchange flows, and concentration risks to support proactive freezing, seizure, and recovery of illicit assets.
Resources	TRM Banking on Stablecoins whitepaper Shadow Bankers – Report TRM Labs

Vendor	
Description	Provides crypto-native intelligence by analyzing blockchain data, exchange activity, and off-chain sources to detect illicit finance across digital-asset ecosystems. By transforming fragmented data into actionable insights, the platform uncovers cross-market manipulation, sanctions evasion, and fraud networks in real time. Its ecosystem-wide intelligence enhances transparency, accelerates investigations, and enables proactive mitigation of financial crime risks for institutions and regulators alike.
Tech used	- REST APIs Provide intelligence through a secure, REST-based API that connects directly to banks', regulators', and Layer-1 ecosystems' existing systems. This modular infrastructure enables seamless integration of digital-asset risk and market surveillance data, allowing

	institutions to act on insights without costly system replacements. - Digital Identity Verification Links identity attributes, sanctions and adverse-media data, and wallet-address patterns across jurisdictions to detect hidden beneficial ownership and high-risk counterparties. This cross-domain intelligence helps prevent infiltration by money-mule networks and improves counterparty due diligence for banks and Layer-1 ecosystems. - Blockchain Transaction Monitoring racks flows and patterns of digital-asset transfers across chains, bridges, and exchanges to identify illicit activity, sanctions evasion, and manipulation. Incorporates market-surveillance analytics—such as order-flow analysis, wash-trading detection, and liquidity-behavior monitoring—to assess market integrity and flag predatory trading patterns that threaten ecosystem stability. - Privacy-Enhancing Tools Maintains complete audit trails of data sources, logic paths, and model outputs, ensuring every decision and risk alert is traceable and defensible. This transparency eliminates black-box opacity and supports regulators’ and institutions’ ability to validate intelligence outcomes while safeguarding sensitive data within controlled environments.
Resources	How Inca Digital Enables Banks to Navigate the Digital Asset Ecosystem How Inca Enables Regulators to Better Understand the Digital Asset Ecosystems

[How Inca Supports Layer1s in Identifying and Managing Risks](#)



Protect

Vendor	 Fireblocks
Description	Provides a secure digital asset custody and transfer platform that protects institutions against hackers, theft and misuse of cryptocurrencies. Combining AI-driven analytics, automated screening, and real-time blockchain monitoring, it detects and blocks illicit flows before settlement. Integrated KYC/KYB/KYT and sanctions controls ensure AML/CFT and FATF Travel Rule compliance, while MPC security and audit trails maintain transparency across stablecoin, tokenized deposit, and CBDC transactions
Tech used	- AI Uses AI-assisted risk and transaction analytics to identify suspicious wallet behavior, flag anomalies in token flows, and monitor for potential sanctions or fraud exposure. Machine learning models continuously learn from transaction outcomes and compliance feedback, reducing false positives and improving real-time threat detection

	- API Provides API-based orchestration that connects wallet infrastructure, blockchain nodes, and compliance providers. These APIs enable automated screening, mint/redeem monitoring, and risk scoring before transactions are executed, ensuring compliant settlement across stablecoin, CBDC, and tokenized deposit systems - Digital Identity Verification Combines KYC/KYB and Know Your Transaction (KYT) protocols to prevent interactions with sanctioned or high-risk entities. Transactions are automatically screened using policy-based rules and data from Chainalysis and Elliptic, covering both inbound and outbound flows to ensure AML/CFT compliance. Supports FATF Travel Rule requirements by enabling cross-jurisdictional data sharing and verification - Blockchain Transaction Monitoring Delivers blockchain-native intelligence that tracks asset movements, and real-time risk checks detect suspicious activity to safeguard digital assets. - Privacy-Enhancing Tools Implements defense-in-depth security through multi-party computation (MPC) and hardware isolation to secure wallet keys and transaction authorizations. Each transaction is processed within a dedicated workspace, ensuring strict segregation of roles and policy-based approvals. Supports tamper-resistant audit trails and encrypted data exchange to maintain transparency while minimizing data exposure.
Resources	How Fireblocks is Powering CBDC and Stablecoin Innovation Through Project Acacia

[Navigating DORA Compliance for Third-Party ICT Vendors](#)

Vendor	 HYPERNATIVE
Description	Helps to identify fraud and illicit activity by analyzing both on-chain and off-chain data in real time, enabling early stage identification of threats before exploits or scams materialize. The platform helps to stop crypto payment fraud/phishing/scamming and security attacks on crypto wallets and smart contracts.
Tech used	- AI Models analyze on-chain data to detect behavioral patterns of malicious activity, while real-time off-chain data integration identifies potential hacks or fraud networks, and correlation of both data sources enables proactive prevention before damage occurs, allowing institutions to act before exploits are finalized on-chain. Machine-learning and graph-based analytics continuously refine risk signals across 300+ threat types. - API Delivers SDKs and REST APIs enabling pre-transaction screening and policy enforcement for payment providers, on/off-ramp operators, and stablecoin issuers. These APIs connect seamlessly to settlement workflows, allowing real-time blocking, whitelisting, or alerting when transactions or addresses exceed configured risk thresholds.

	- Digital Identity Verification Provides real-time address reputation scoring and counterparty risk assessment across chains. Integrates sanctions, mixer, and exploit-address data to vet wallet ownership and behavioral patterns, preventing engagement with known illicit entities while ensuring compliant digital-asset payments. - Blockchain Transaction Monitoring Monitors activity across multi-hop and cross-chain transactions, bridges, and liquidity pools to detect layering, wash trading, or illicit fund movement. Supports visualization of fund flows and enables pre-block intervention, helping institutions prevent stablecoin misuse or stolen-asset transfer before confirmation. -Privacy-Enhancing Tools Implements privacy-preserving, on-node monitoring that filters non-compliant blocks in real time based on OFAC, sanctions, mixer, and fraud address data. Compliance lists and policy thresholds are fully customizable, enabling institutions to tailor enforcement without compromising validator performance or transaction speed. This framework ensures regulatory adherence, operational efficiency, and minimal data exposure.
Resources	Dismantling the Fraud Barrier to Stablecoin Payments Adoption Hypernative Hypernative Screener Screens Addresses & Identifies Risks Payment Providers Financial Institutions Staking Monitoring